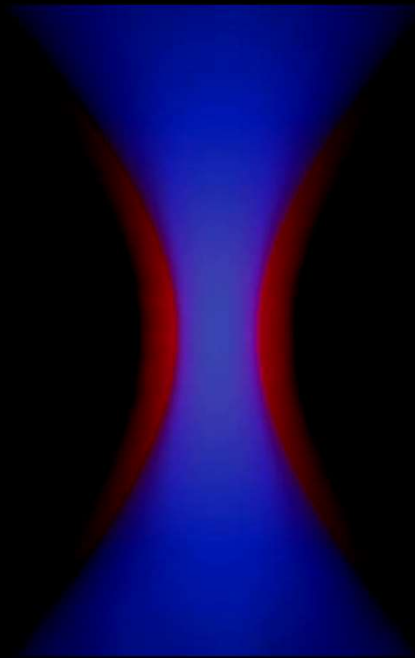




FLUCTUATION MICROSCOPIES

Emmanuel MARGEAT
CNRS

Centre de Biologie Structurale - Montpellier

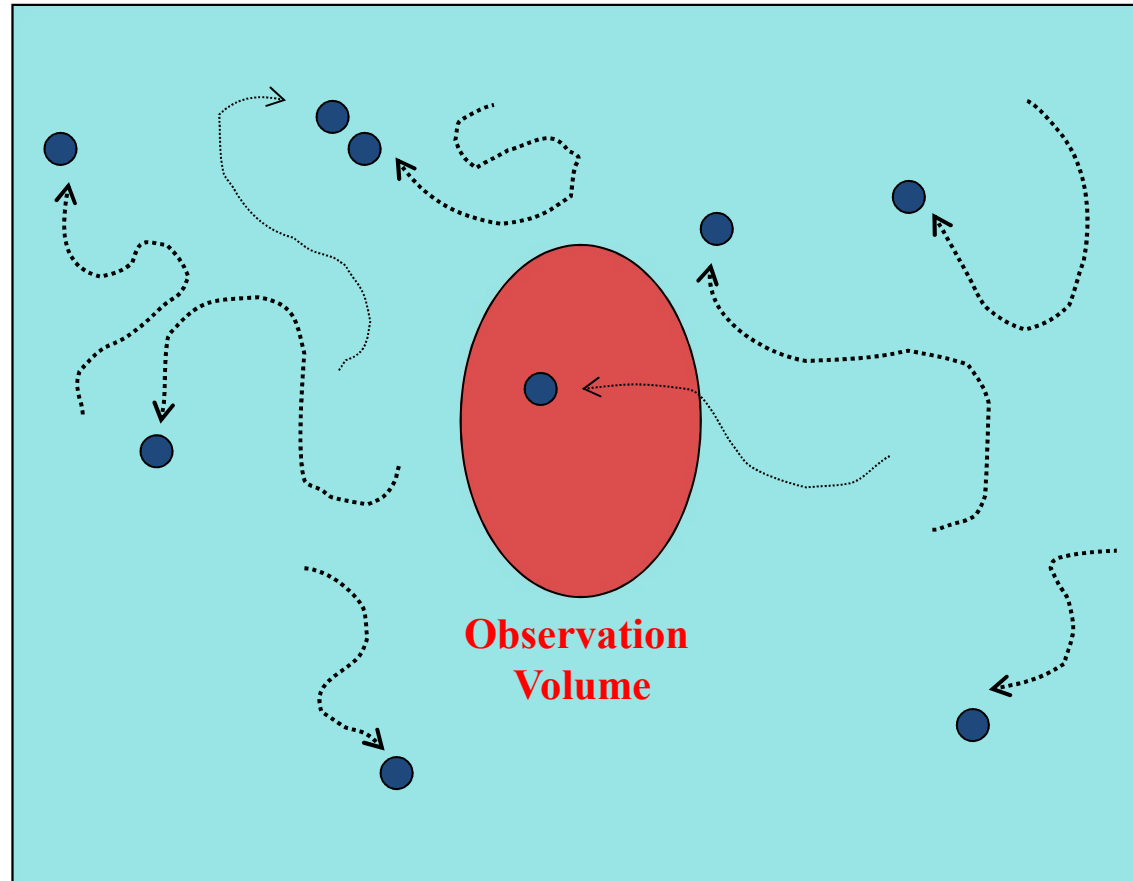
FLUCTUATION MICROSCOPIES

- INTRODUCTION
- PRINCIPE OF FCS
- AUTOCORRELATION
- CROSS CORRELATION
- EXPERIMENTAL DETAILS
- NUMBER & BRIGHTNESS

FLUCTUATION MICROSCOPIES

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- NUMBER & BRIGHTNESS

Observation of fluctuations



Sample Space

What can we observe ?

1. Speed of the movement

2. Particles concentration

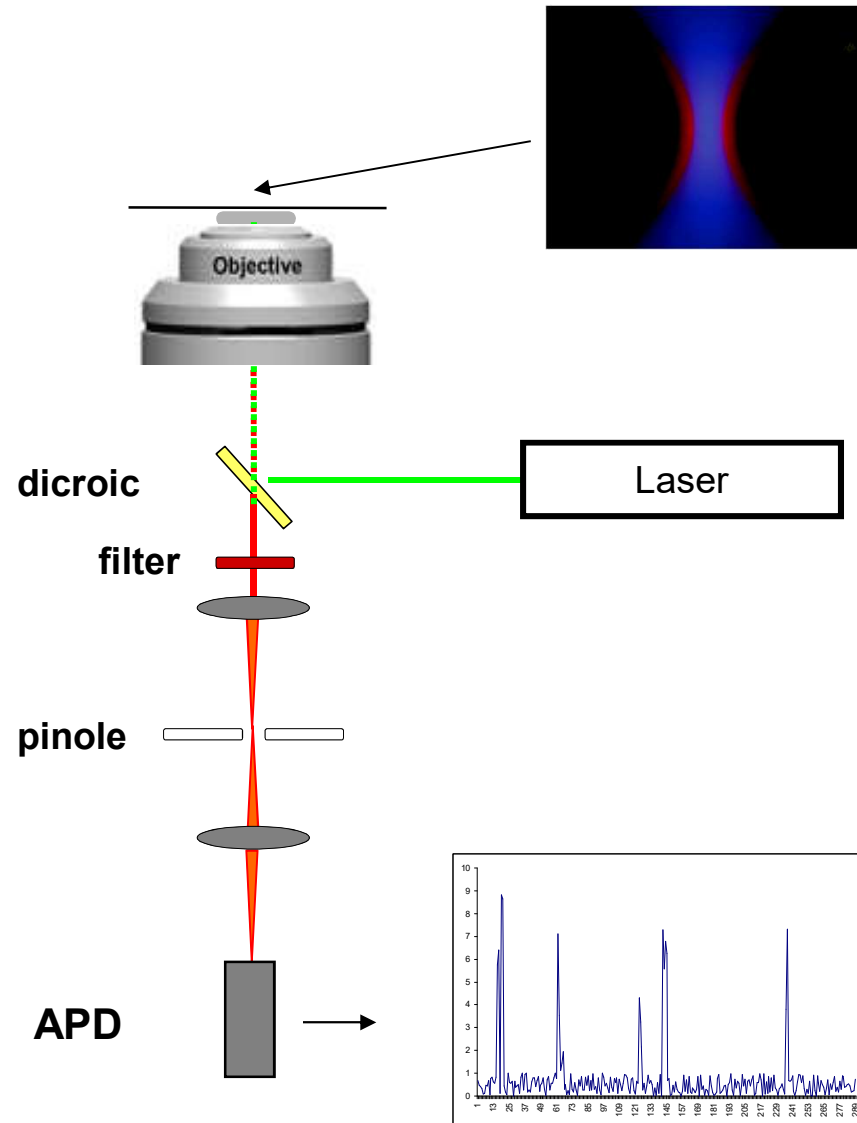
**3. Change of state, and thus
of the signal, of the
particles**

→ Particles interactions

FLUCTUATION MICROSCOPIES

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- CROSS CORRELATION
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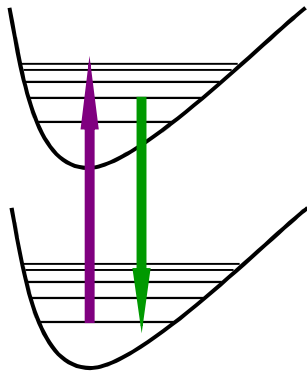
Observing fluctuations : fluorescence microscopy



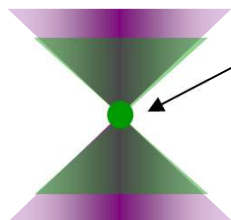
The observation volume

One- & Two-Photon Excitation.

1 - Photon



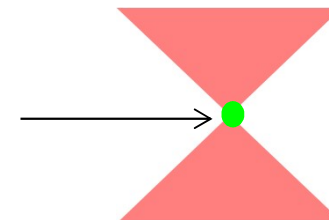
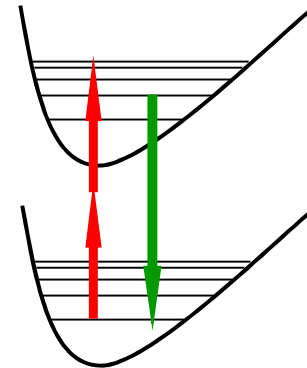
Defined by the size of the pinole, the wavelength, the magnification, and the numerical aperture



$\sim 1 \text{ } \mu\text{m}^3$

Defined by the wavelength and the numerical aperture

2 - Photon



1-photon

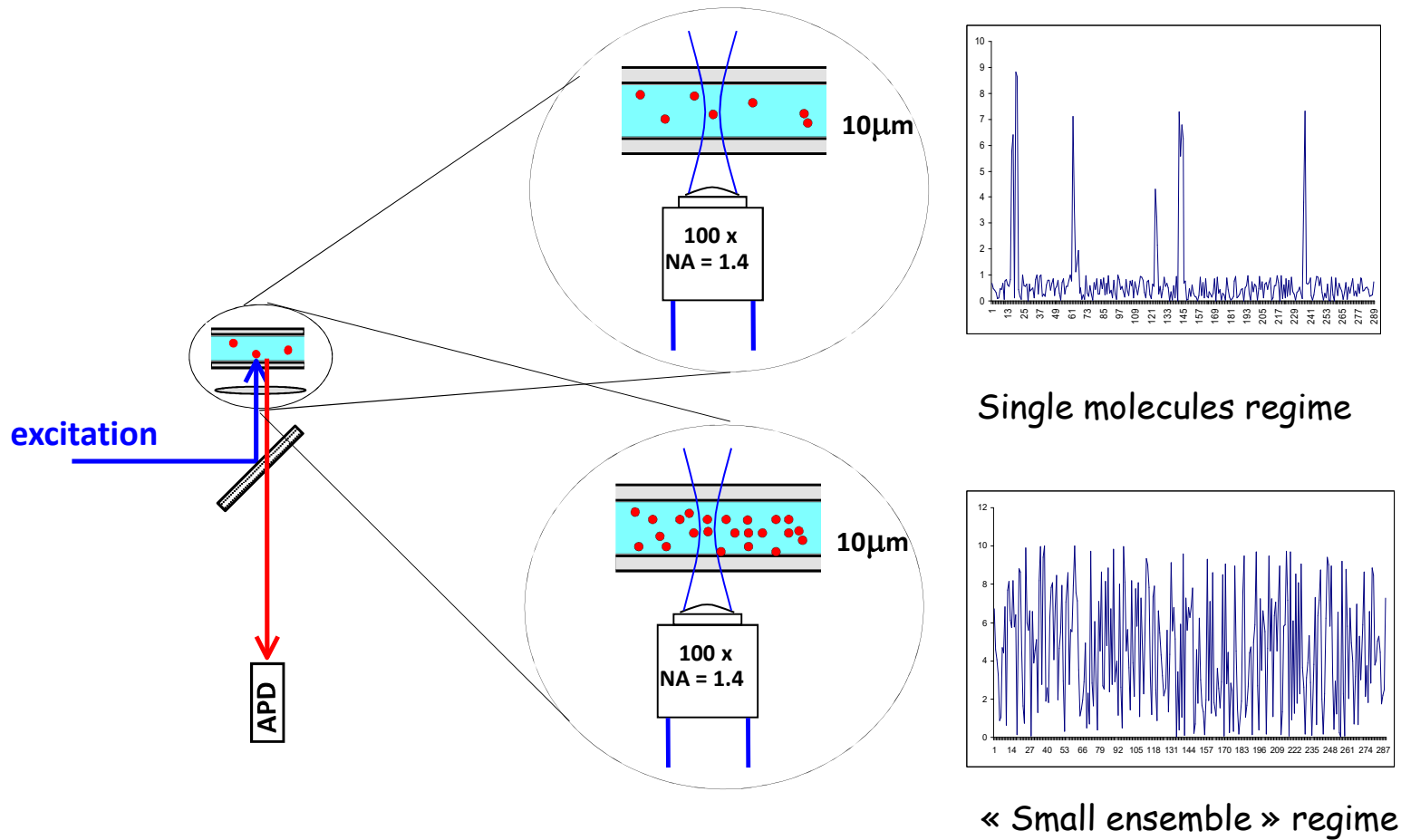
Need a pinhole to
define a small volume



2-photon

Brad Amos
MRC, Cambridge, UK

Fluorescence correlation spectroscopy

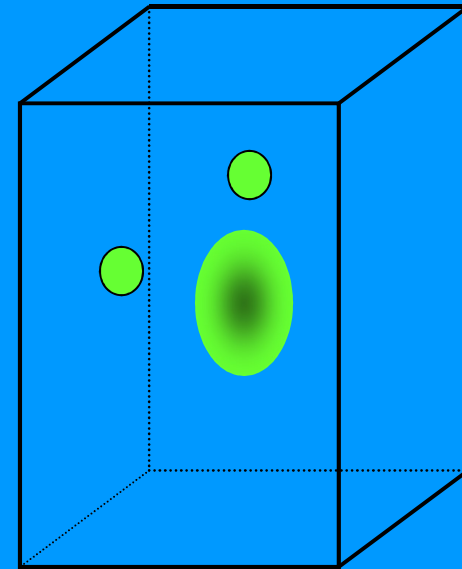


Using FCS, we will analyze these intensity fluctuations

FLUCTUATION MICROSCOPIES

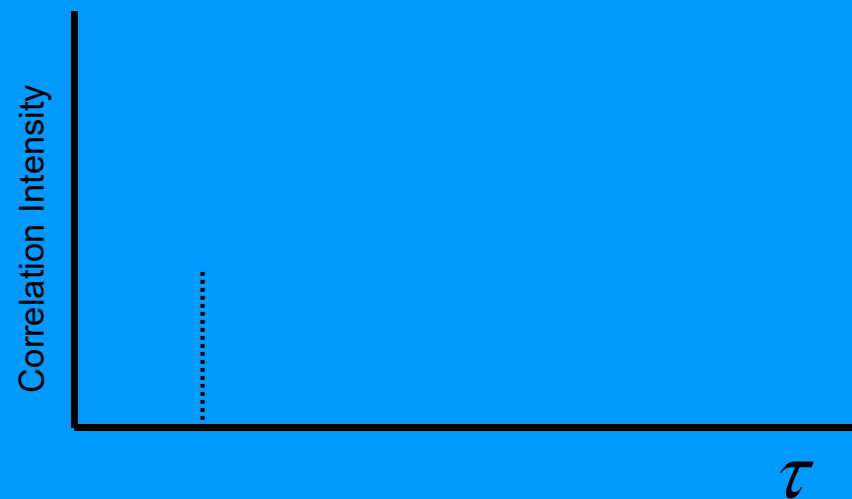
- INTRODUCTION
- PRINCIPE OF FCS
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Introduction to correlation function.

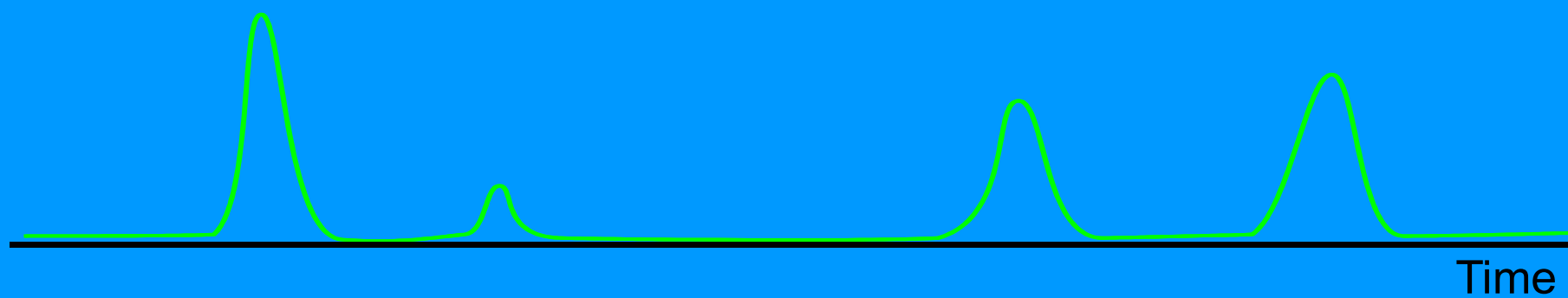


Time

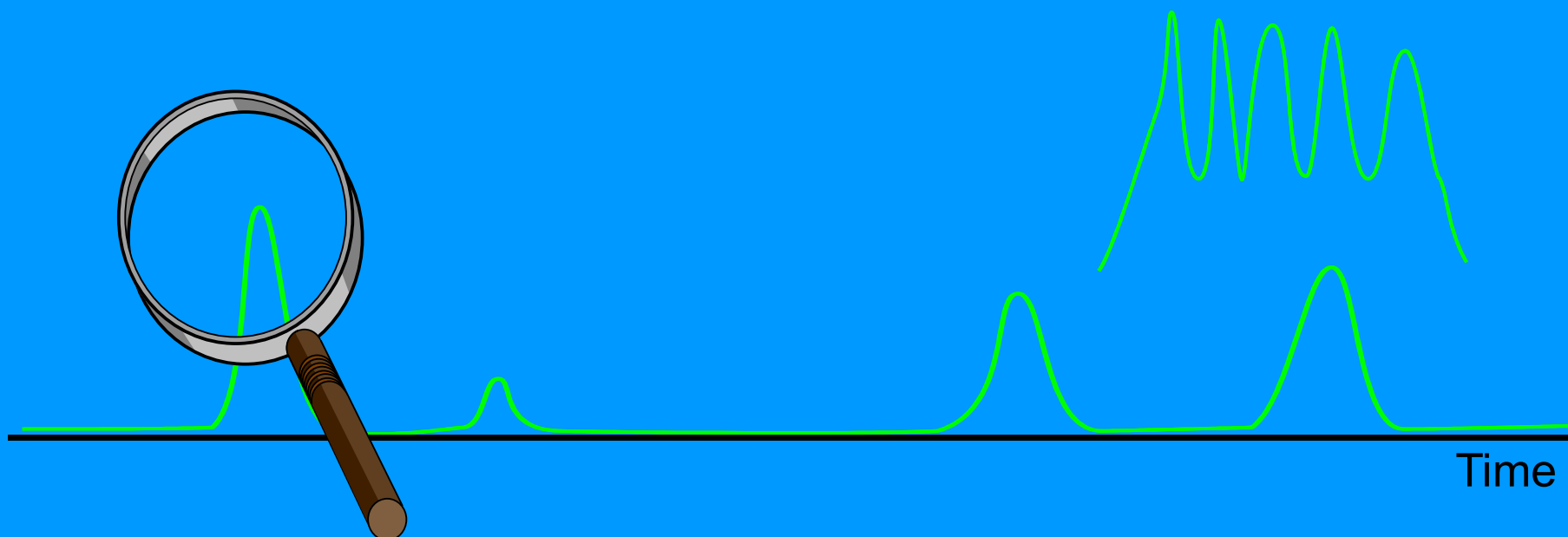
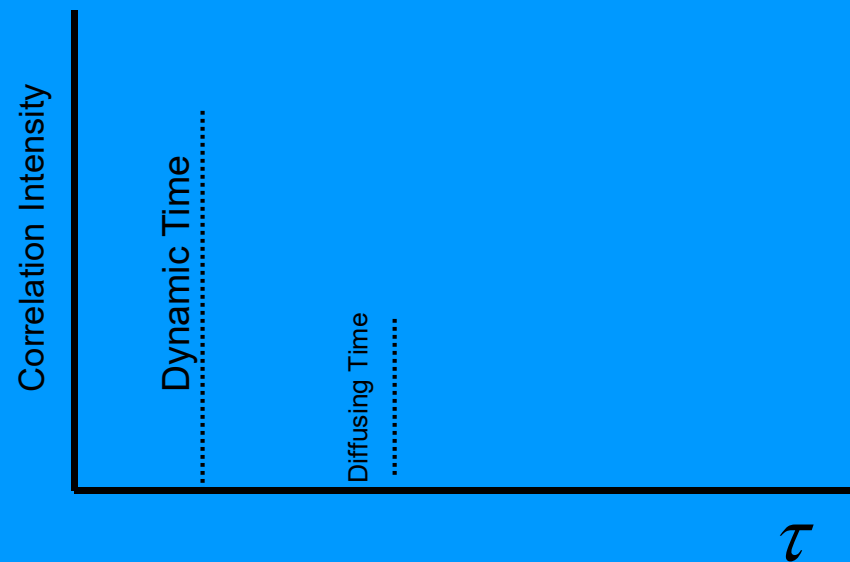
AutoCorrelation. Diffusion only, No Dynamics. “Donor-Donor”



$$G(\tau) = \frac{\langle F(t + \tau) \cdot F(t) \rangle}{\langle F \rangle^2}$$



AutoCorrelation. Diffusion and Dynamics. "Donor-Donor"



2 definitions for the autocorrelation function

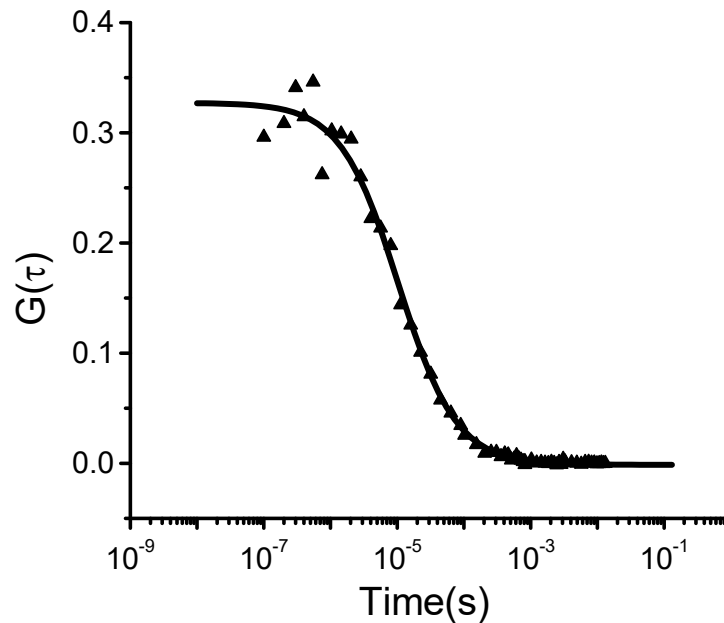
$$G(\tau) = \frac{\langle \delta F(t) \delta F(t + \tau) \rangle}{\langle F \rangle^2}$$

$$\delta F(t) = F(t) - \langle F(t) \rangle$$

Decays to 0 for $\tau \rightarrow \infty$

$$G(\tau) = \frac{\langle F(t + \tau) \cdot F(t) \rangle}{\langle F \rangle^2}$$

Decays to 1 for $\tau \rightarrow \infty$



$$G(\tau) = \frac{1}{N} \left(1 + \frac{\tau}{\tau_d} \right)^{-1} \left(1 + \frac{r_o^2 \tau}{z_o^2 \tau_d} \right)^{-1/2}$$

Where r_o and z_o are the lateral and axial radii of the observation volume respectively

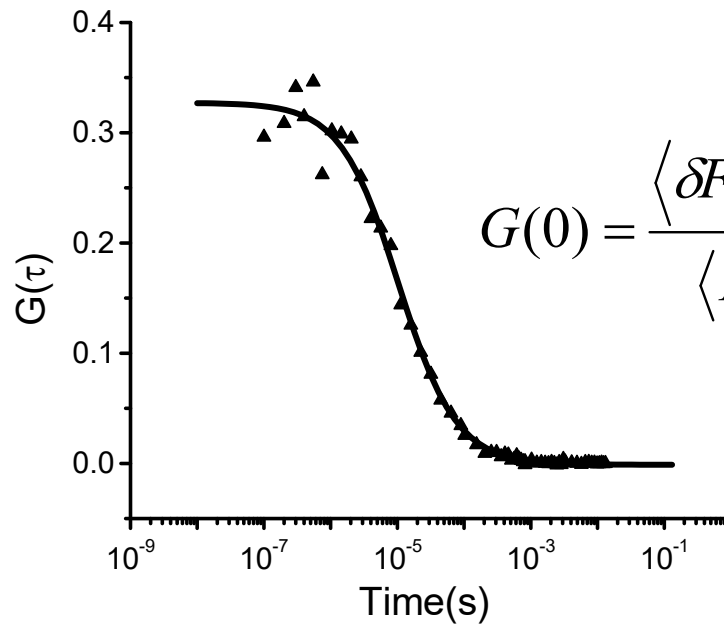
$G(0) = 1/N$ of observed particles

In a poissonian system the variance of the number of events is equal to its mean
(here the average number of observed particles)

$$\langle N \rangle = \text{Variance}_N = \sigma_N^2$$

$F = \varepsilon.N$, avec ε = molecular brightness

$$G(\tau) = \frac{\langle \delta F(t) \delta F(t + \tau) \rangle}{\langle F(t) \rangle^2}$$



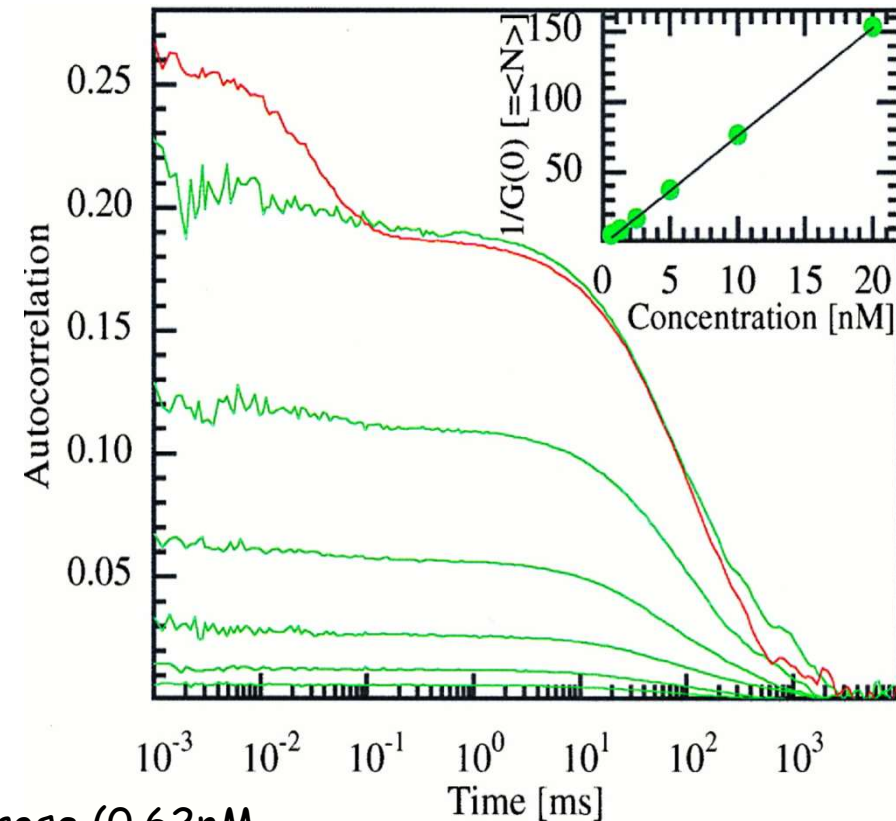
$$G(0) = \frac{\langle \delta F(t)^2 \rangle}{\langle F \rangle^2} = \frac{\langle (F(t) - \langle F(t) \rangle)^2 \rangle}{\langle F \rangle^2} = \frac{\sigma_F^2}{\langle F \rangle^2} = \frac{\varepsilon^2 \cdot \sigma_N^2}{\varepsilon^2 \cdot \langle N \rangle^2}$$

$$G(0) = \frac{\text{Variance}}{\langle N \rangle^2} = \frac{1}{\langle N \rangle}$$

Effect of the concentration

For a single species, no photobleaching

$$G(0) = \frac{1}{\langle N \rangle}$$



Rhodamine 6G in 70% sucrose (0.62nM,
1.25nM, 2.5nM, 5nM, 10nM, 20nM)
(green curves)

0.62nM with a 70 times higher
excitation power (red curve)

The Effects of Particle Size on the Autocorrelation Curve

Diffusion Constants

300 $\mu\text{m}^2/\text{s}$

90 $\mu\text{m}^2/\text{s}$

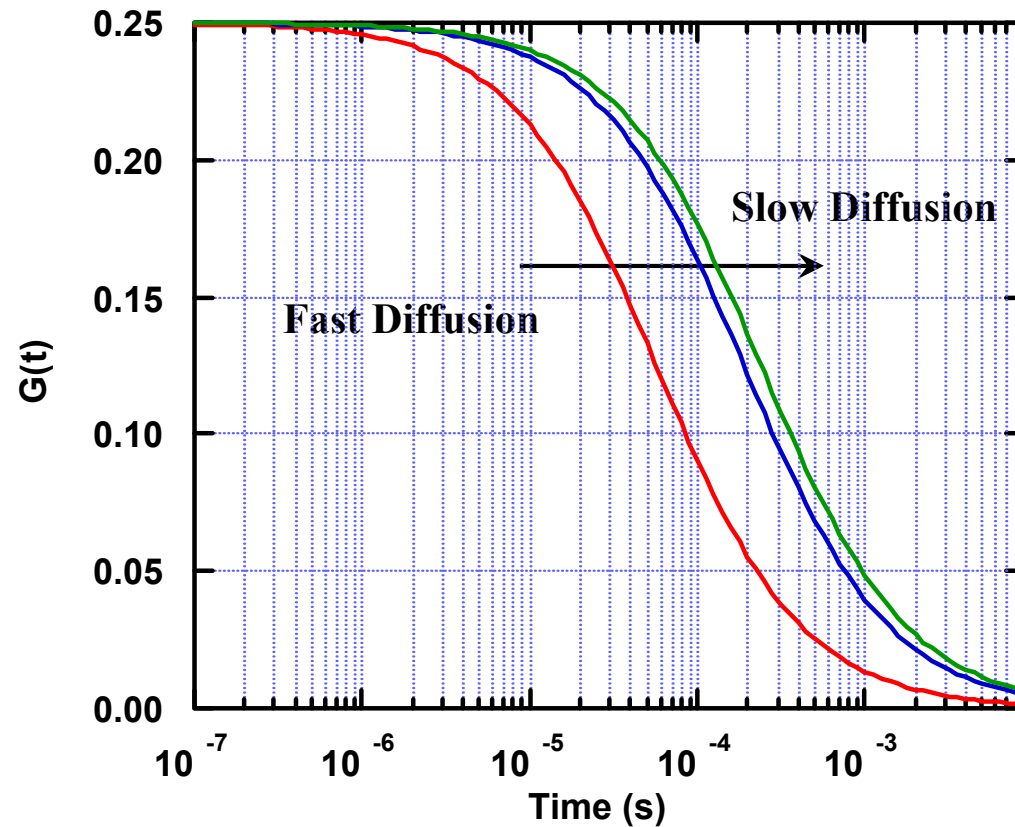
71 $\mu\text{m}^2/\text{s}$

Stokes-Einstein Equation:

$$D = \frac{k \cdot T}{6 \cdot \pi \cdot \eta \cdot r}$$

and

$$MW \propto \text{Volume} \propto r^3$$



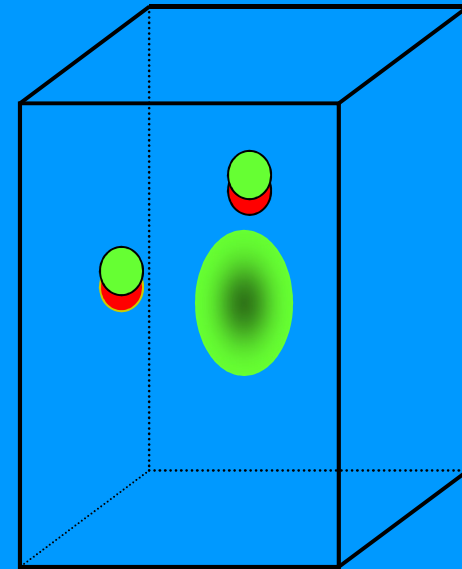
$$\tau_d = \frac{r_o^2}{4D} \quad \text{and} \quad \tau_d = \frac{r_o^2}{8D}$$

for one and two photon excitation respectively

FLUCTUATION MICROSCOPIES

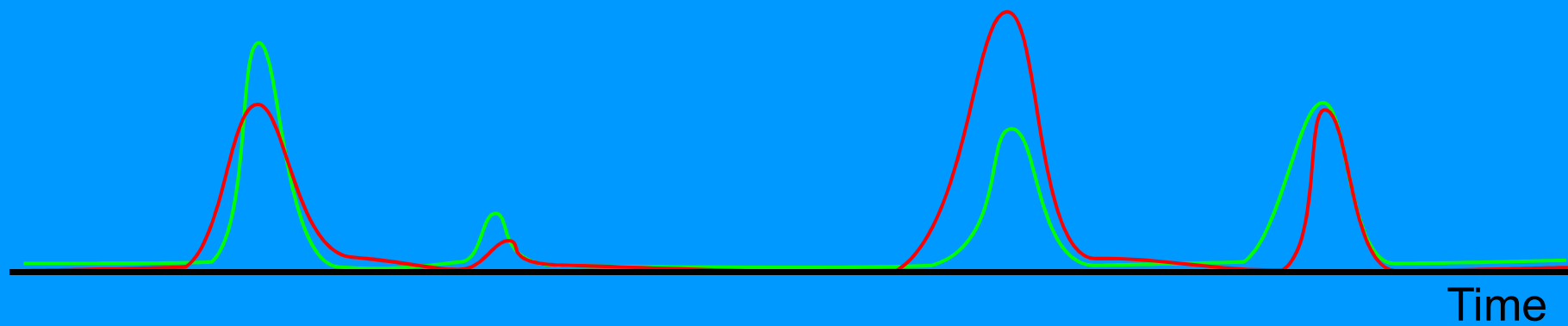
- INTRODUCTION
- PRINCIPE OF FCS
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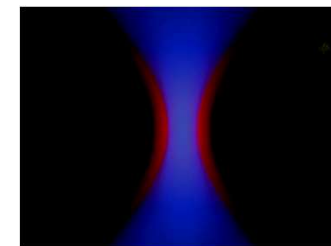
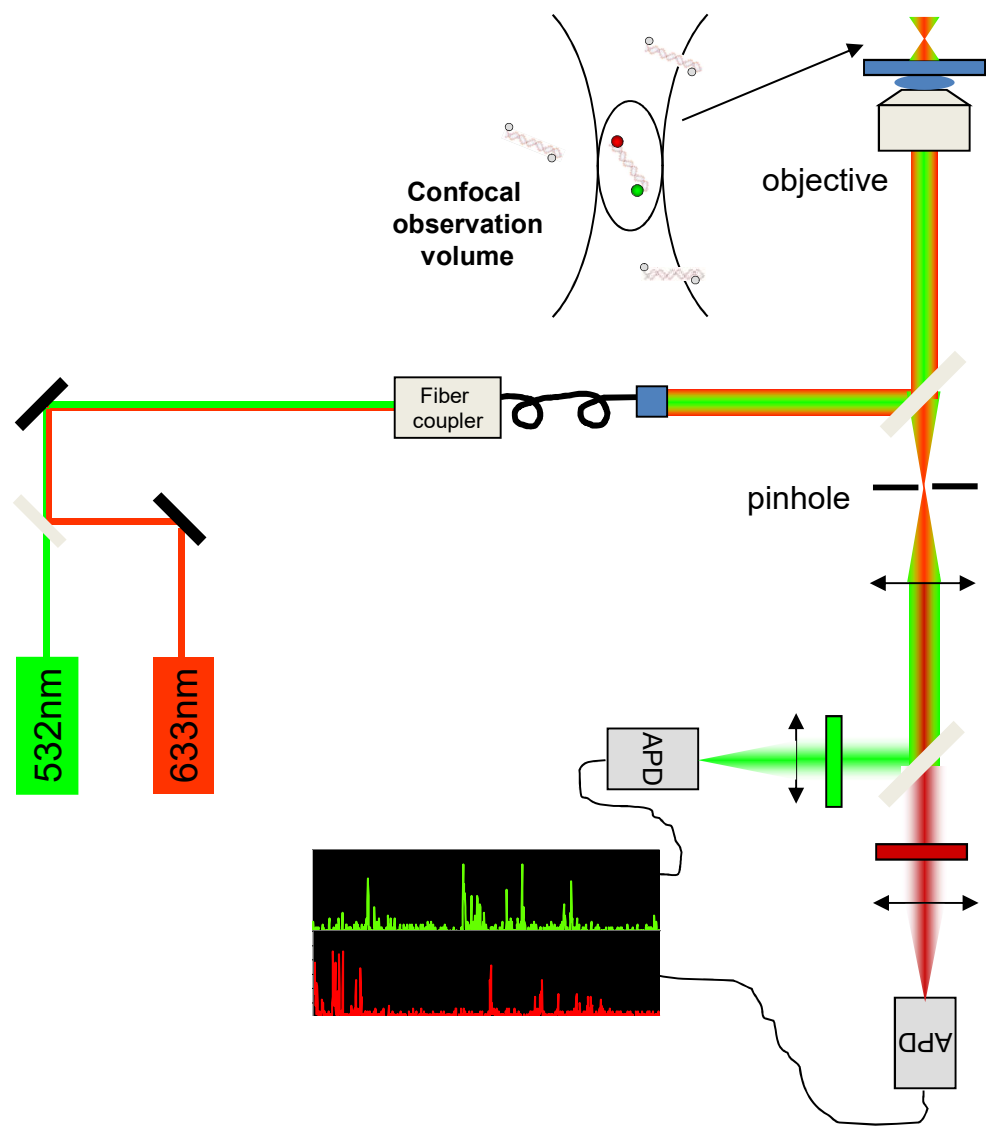
Introduction to correlation function.



Time

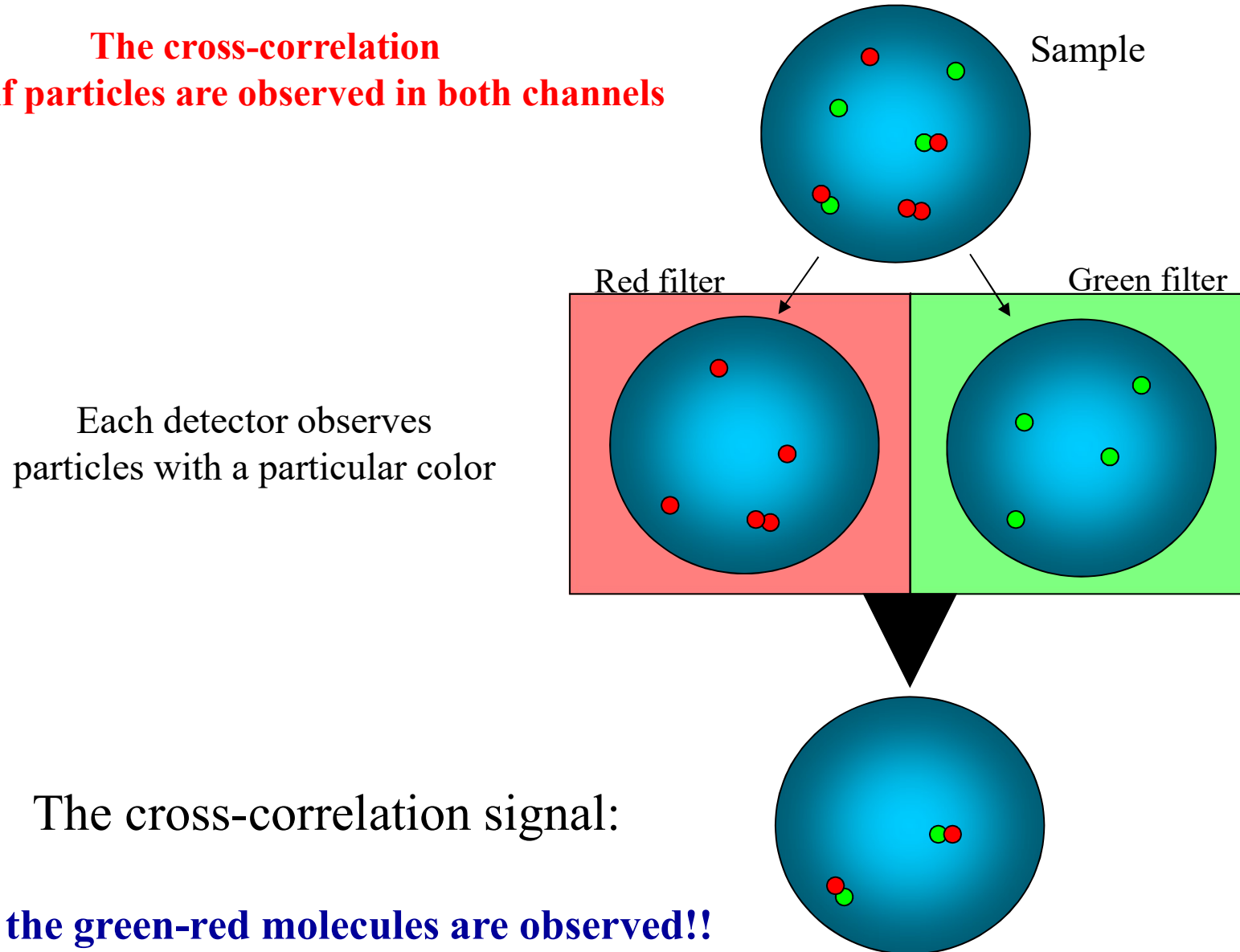
CrossCorrelation. Diffusion only, No Dynamics. “Donor-Acceptor”



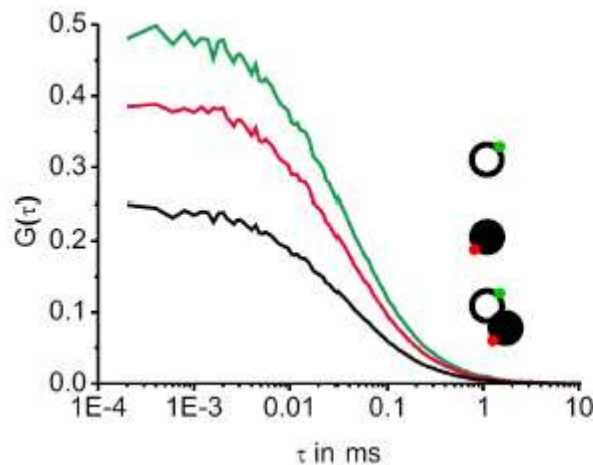


Two-Color Cross-correlation

**The cross-correlation
ONLY if particles are observed in both channels**



$$G_X(\tau) = \frac{\langle \delta F_1(t) \cdot \delta F_2(t + \tau) \rangle}{\langle F_1(t) \rangle \langle F_2(t) \rangle}$$



- n_1 : number of green molecules
- n_2 : number of red molecules
- n_{12} : number of green-red molecules

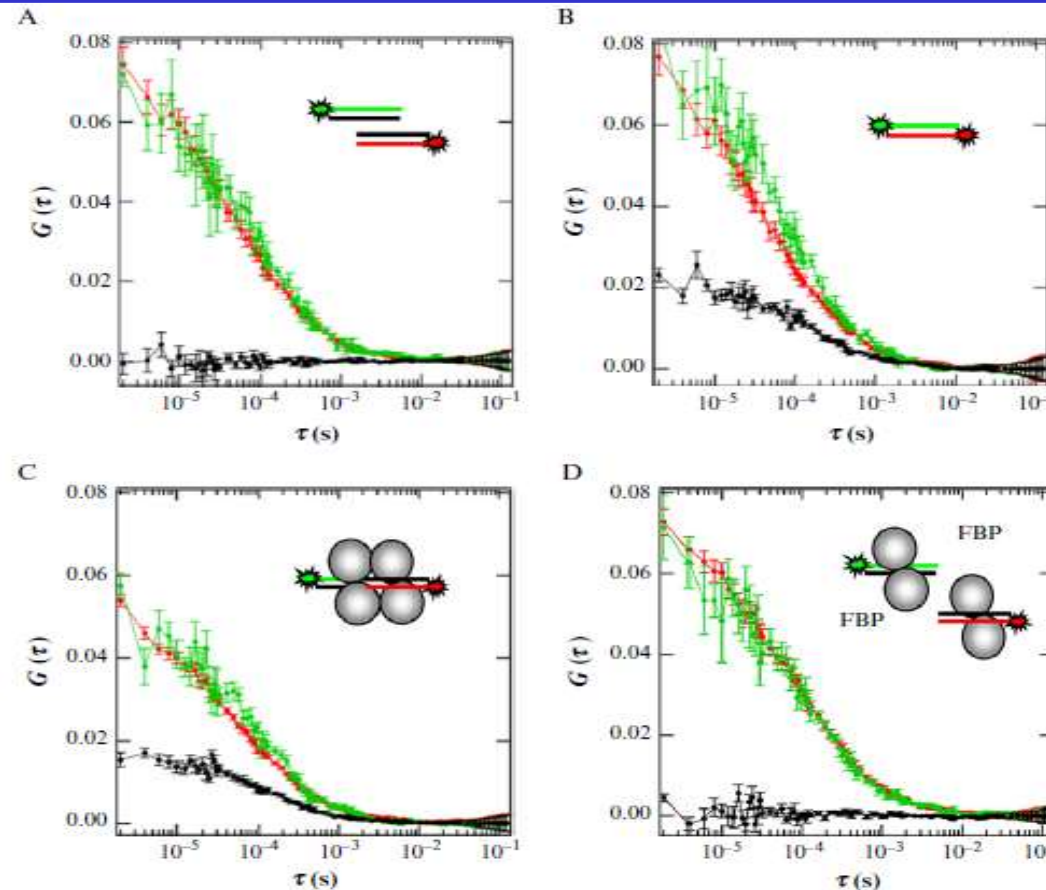
$$g_1(0) = \frac{1}{n_1 + n_{12}}, \quad g_2(0) = \frac{1}{n_2 + n_{12}}, \quad g_{\times}(0) = \frac{n_{12}}{(n_1 + n_{12})(n_2 + n_{12})}$$

$$\frac{g_{\times}(0)}{g_2(0)} = \frac{n_{12}}{n_1 + n_{12}} \rightarrow \text{fraction of green in complex with red}$$

$$\frac{g_{\times}(0)}{g_1(0)} = \frac{n_{12}}{n_2 + n_{12}} \rightarrow \text{fraction of red in complex in complex with green}$$

In vitro interaction between the *CGGR* repressor and its operator

Nathalie Declerck (CBS)



FBP induces the dissociation of the *CGGR* tetramer from its operator

FBP: fructose-1,6-bis-phosphate

FLUCTUATION MICROSCOPIES

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Experimental details

- Background
- Photobleaching
- Size and shape of the observation volume
- Cross-talk between detection channels (FCCS)
- Afterpulsing

Background

- Constant, uncorrelated (B)
→ Diminishes the value of $G(0)$

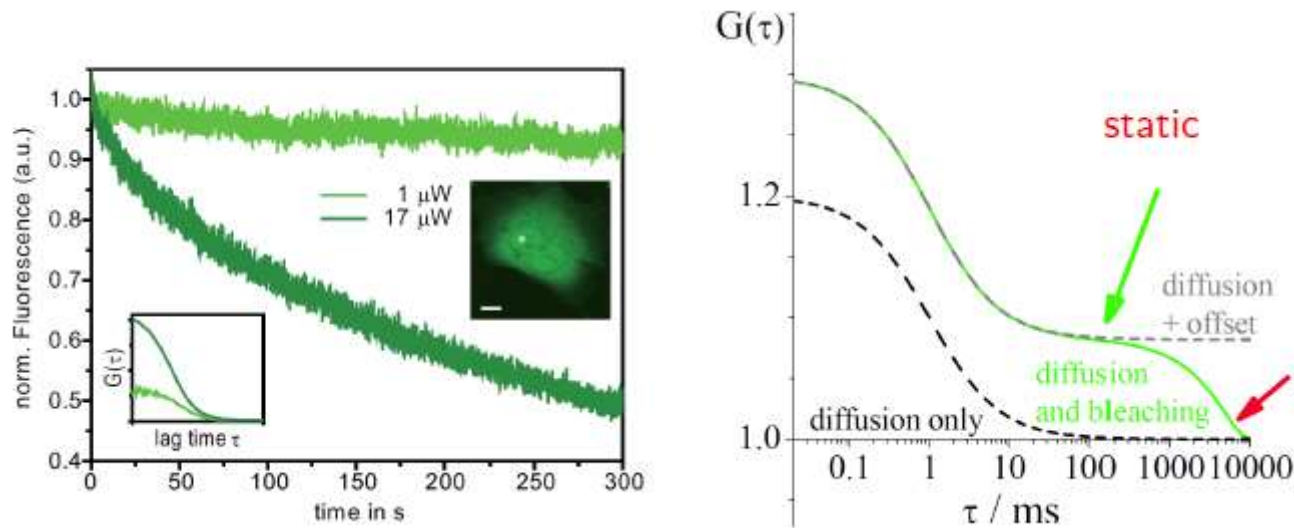
$$g'(\tau) = g(\tau) \left(\frac{\langle F \rangle}{\langle F \rangle + B} \right)^2$$

- Correlated (arising from diffusing molecules)
→ Add a new component to the analysis

$$G(\tau) = \sum_{i=1}^M f_i^2 \cdot G(0)_i \cdot \left(1 + \frac{8D\tau}{w_{2DG}^2} \right)^{-1} \quad (2D\text{-Gaussian, 2PE})$$

Photobleaching

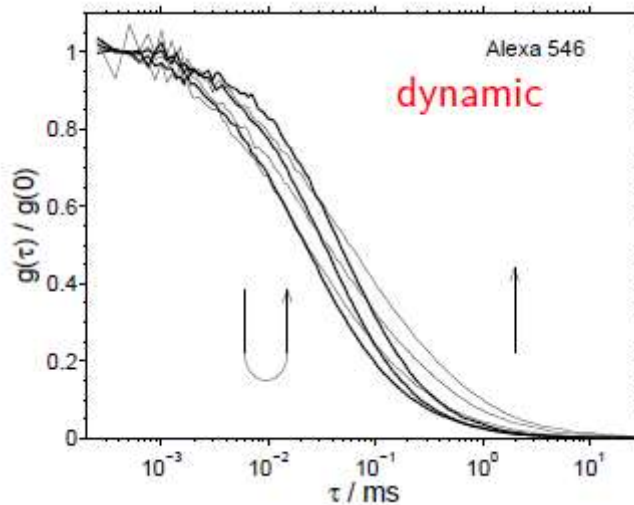
- Static (photobleaching of quasi immobile molecules)



Photobleaching

- Dynamic

→ Diminishes the apparent diffusion coefficient

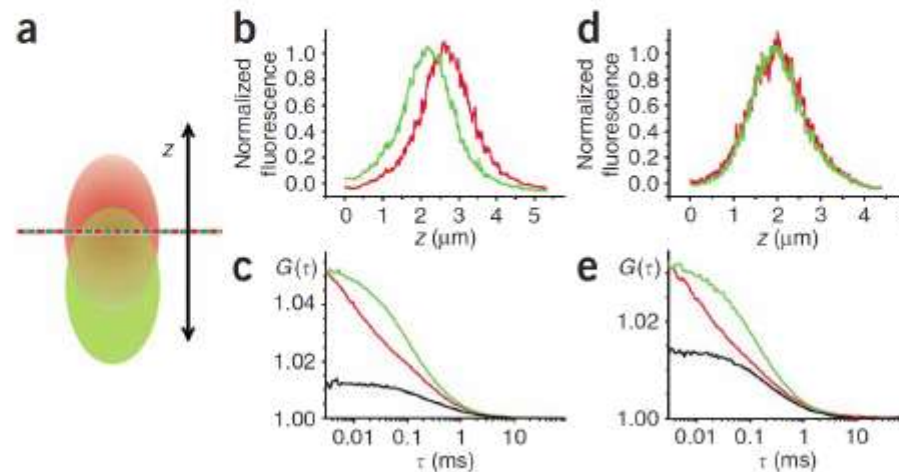


Size and shape of the observation volume

FCS: distortions of the volume size can affect:

- measured diffusion coefficient D via r_0 in $D = r_0^2 / \tau_D$
- measured concentration via changed V_{eff} : $g(0) = 1/cV_{\text{eff}}$

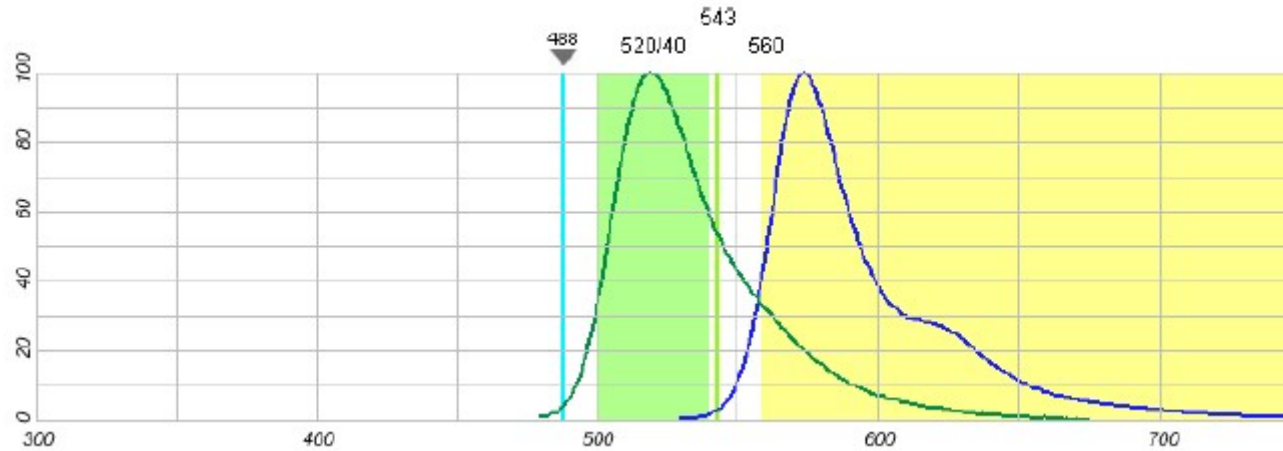
FCCS: in addition to the above, the volume overlap affects $g_{\times}(\tau)$



— use objective correction collar for coverslip thickness correction

NB : Using 2photon excitation, the excitation volumes are identical

Spectral cross talk (FCCS)

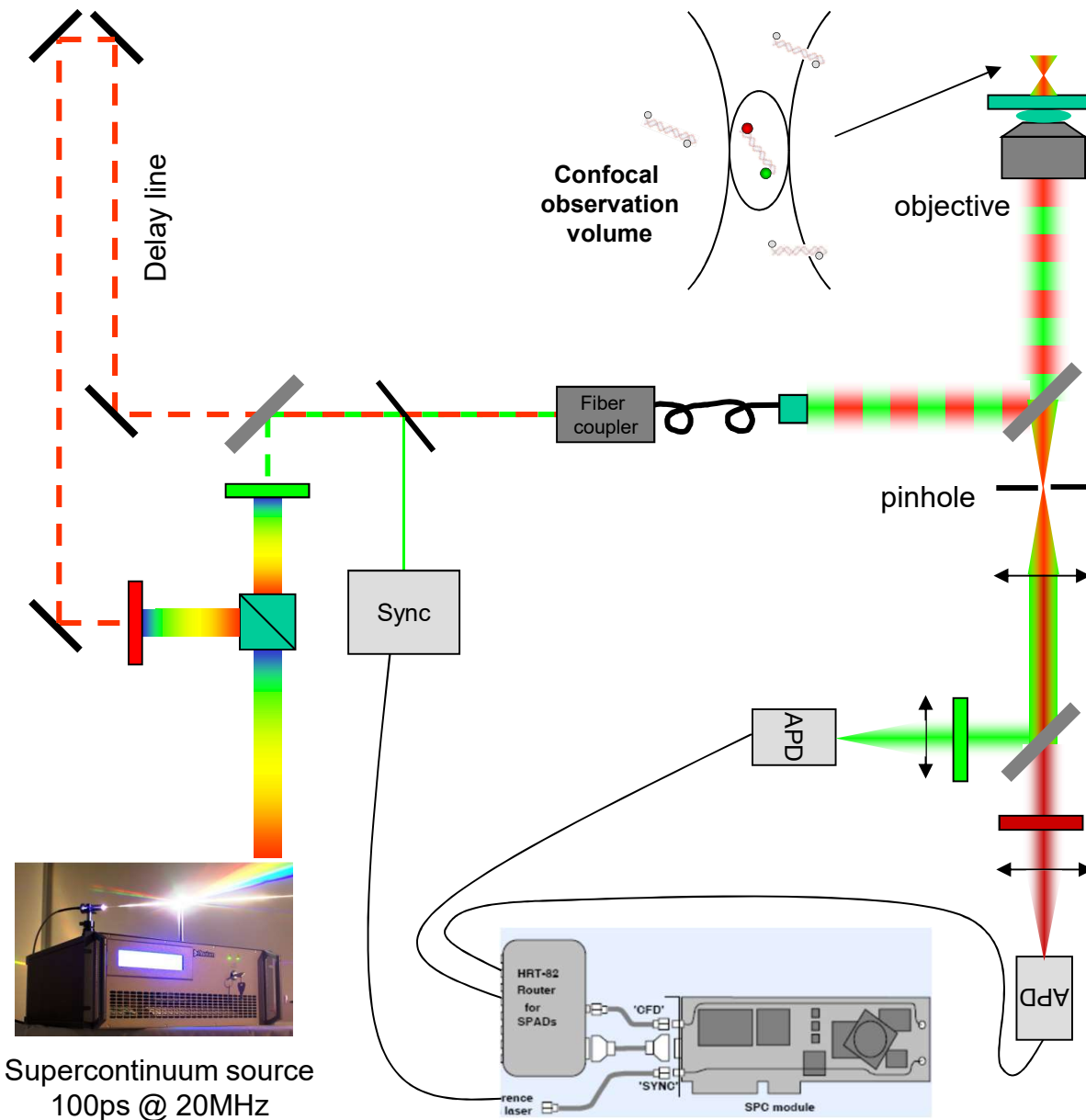


Detection of the « green » fluorophore in the « red » detection channel : → Artifactual cross correlation

- Use spectrally distinct fluorophores
- Use alternated laser excitation (nsALEX / PIE)

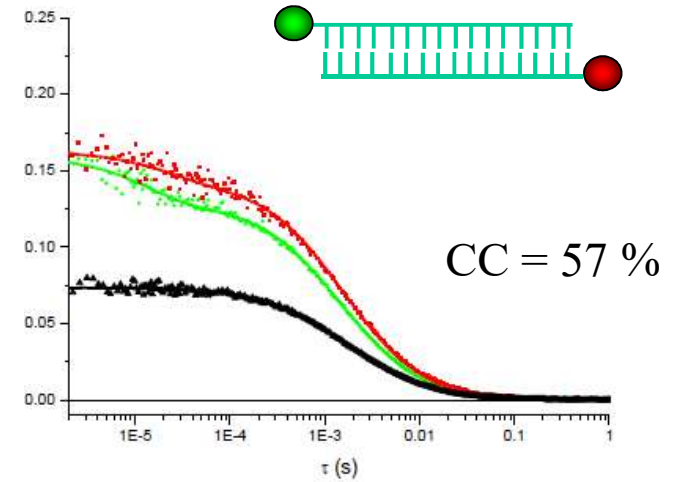
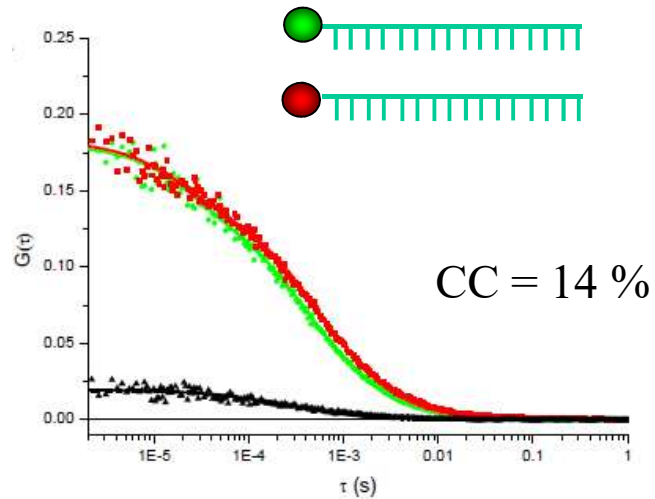
ALEX : Alternating laser excitation, PIE : Pulsed interleaved excitation

PIE / nsALEX with a supercontinuum source

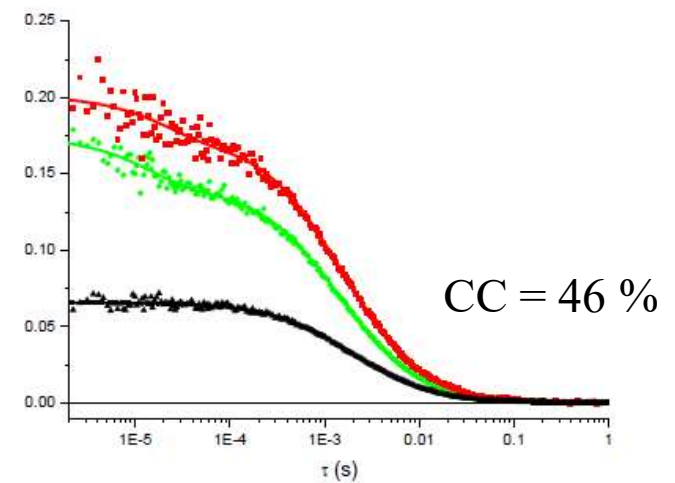
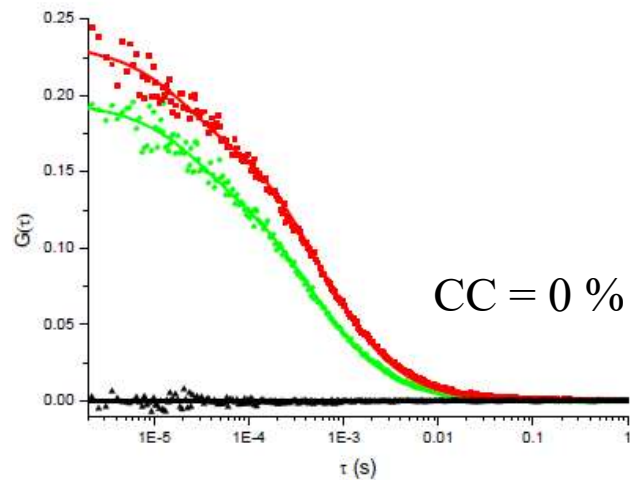


Removal of crosstalk in FCCS

Regular
FCCS

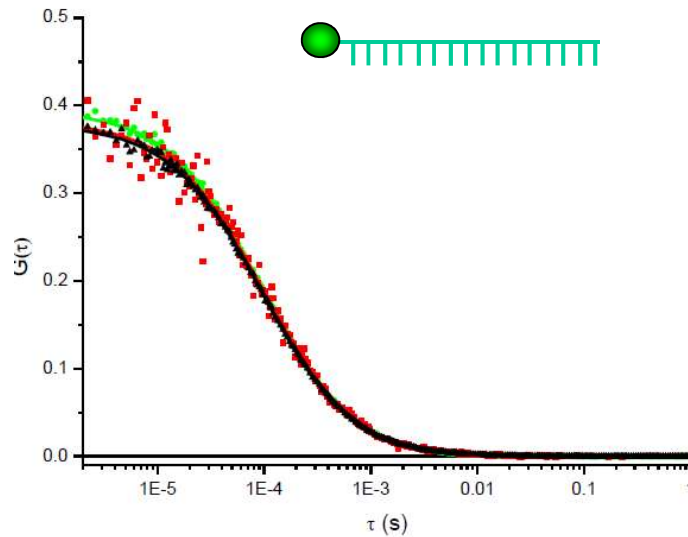


PIE
FCCS



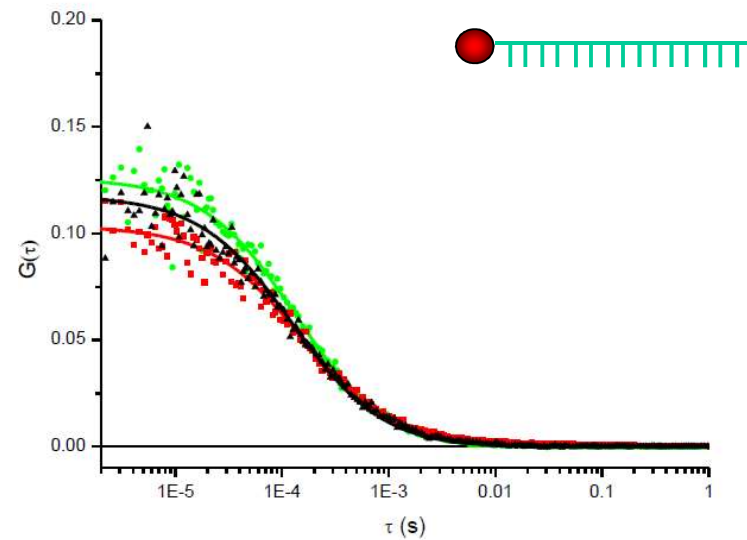
- Green AC
- Red AC
- Green Red CC

Overlap of the excitation and detection volumes



Green
excitation

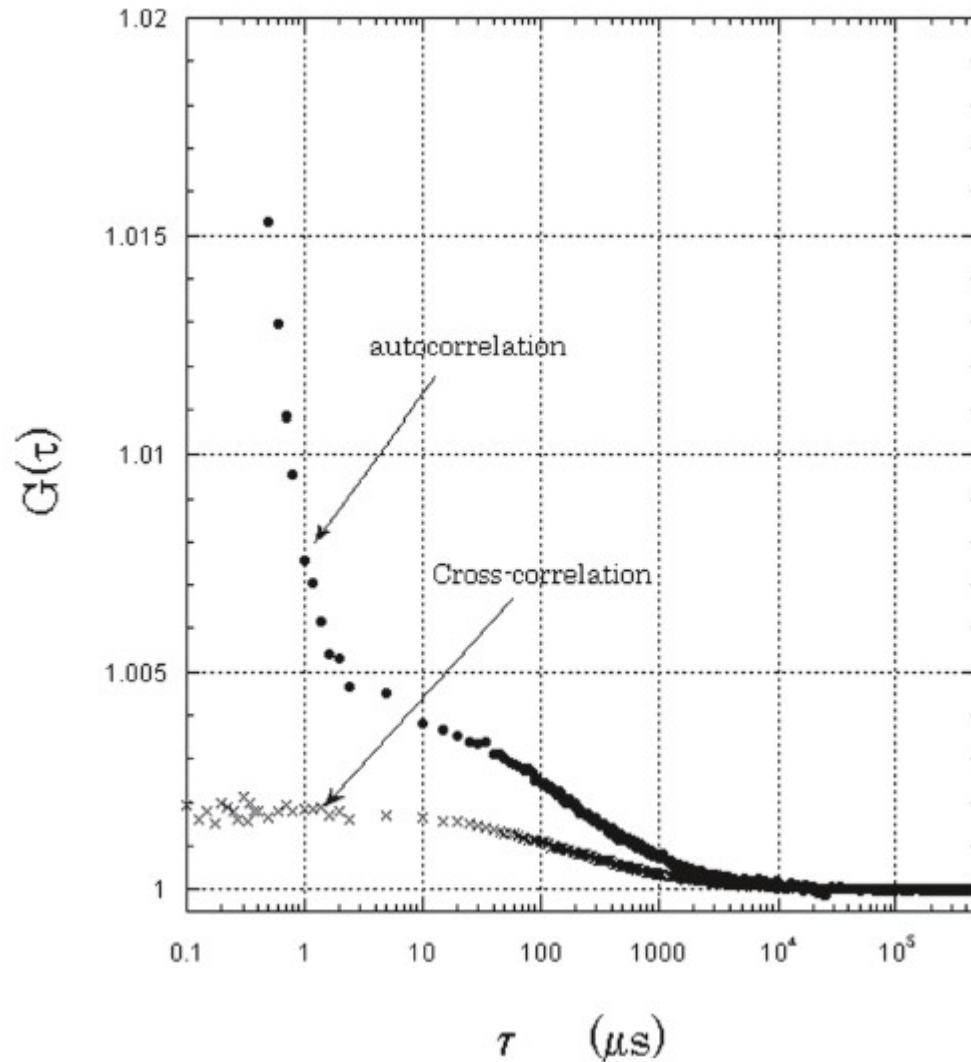
— Green detection AC
— Red detection AC
— Green Red CC



— Green excitation AC
— Red excitation AC
— Green Red CC

Red
detection

Afterpulsing



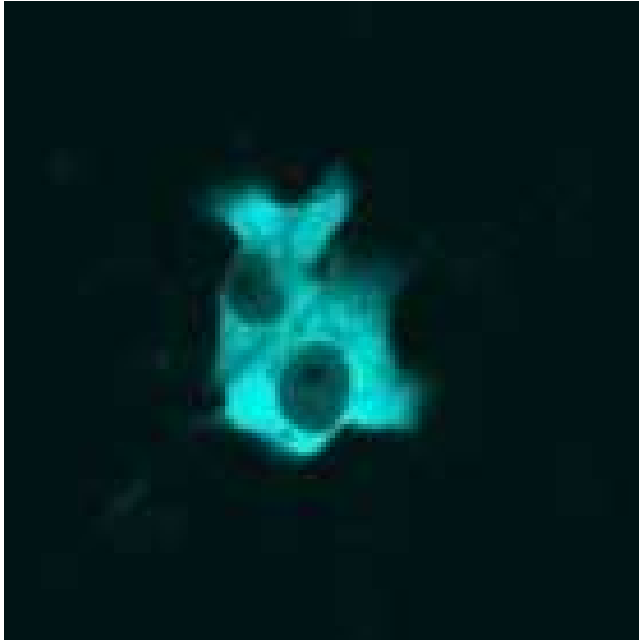
Autocorrelation :
One detector

Cross correlation :
Two spectrally equivalent detectors
50/50 beam splitter

FLUCTUATION MICROSCOPIES

- INTRODUCTION
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FCS and cell biology



An FCS analysis on all pixels of the image is impossible due to

- Photobleaching
- Measurement time

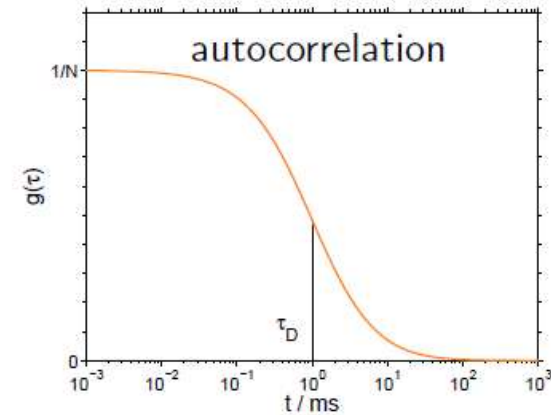
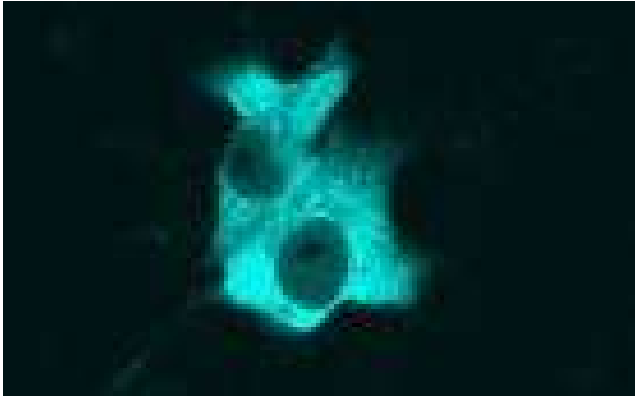
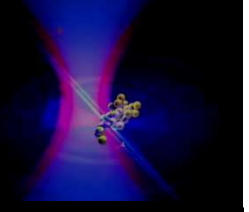
Une mesure et une analyse en *Number & Brightness* (N&B) va nous permettre de mesurer rapidement les paramètres de

- Concentration (N)
- Oligomerisation / brillance (B)

En cela, l'analyse N&B s'apparente au PCH
→ ne permet pas la détermination de D

- Introduction
- N&B theory
- Applications
- Cross correlation

The number and brightness (N&B) analysis



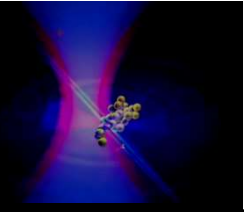
- Building a correlation curve takes time
 - For imaging : 1 curve / pixel
 - Photobleaching becomes rapidly and issue

→ If you are only interested in determining the concentration and brightness of the molecules, this can be done very fast, in scanning / imaging mode

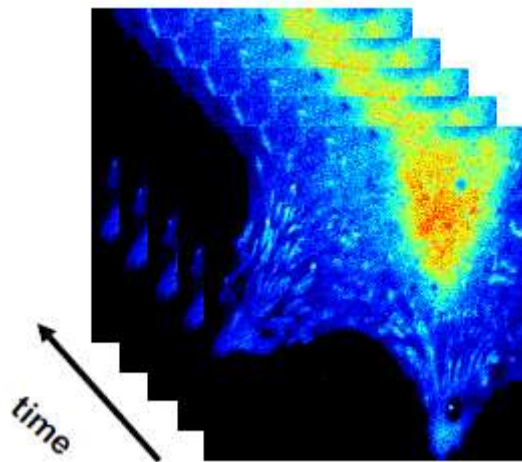
Number and Brightness

- Introduction
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Principle of the N&B analysis



We acquire a stack of K images (at low pixel dwell time) and we define for each pixel :



$$F = \frac{\sum_K F(x, y)}{K}$$

$$\sigma_F^2 = \frac{\sum_K (F(x, y) - F)^2}{K}$$

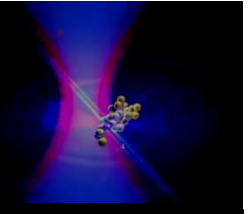
And we can calculate the apparent number and apparent brightness

$$\frac{\sigma_F^2}{\langle F \rangle^2} = \frac{1}{\langle N \rangle} \Rightarrow \langle N \rangle = \frac{\langle F \rangle^2}{\sigma_F^2}$$

$$\langle F \rangle = B \langle N \rangle \Rightarrow B = \frac{\sigma_F^2}{\langle F \rangle}$$

- Introduction
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Principle of the N&B analysis



However, to recover the true number and true brightness we need to take into consideration the various parameters contributing to the variance

n = true number
 ε = true brightness

Total variance of the signal

$$\sigma_F^2 = \sigma_n^2 + \sigma_d^2 = \varepsilon^2 n + \varepsilon n$$

$$\langle F \rangle = \varepsilon n$$

σ_n^2 The variance in the signal due to variation in n scales with the square of the brightness

σ_d^2 The variance in the signal due to the detection process scale with the brightness for a photon counting detector

$$B = \frac{\sigma_F^2}{\langle F \rangle} = \frac{\varepsilon^2 n + \varepsilon n}{\varepsilon n} = 1 + \varepsilon$$

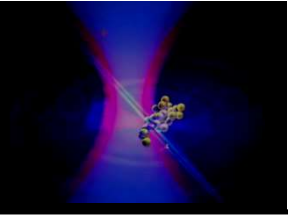
$$\langle N \rangle = \frac{\langle F \rangle^2}{\sigma_F^2} = \frac{\varepsilon^2 n^2}{\varepsilon^2 n + \varepsilon n} = \frac{\varepsilon n}{\varepsilon + 1}$$

$$\varepsilon = \frac{\sigma_F^2 - \langle F \rangle}{\langle F \rangle}$$

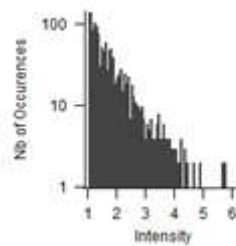
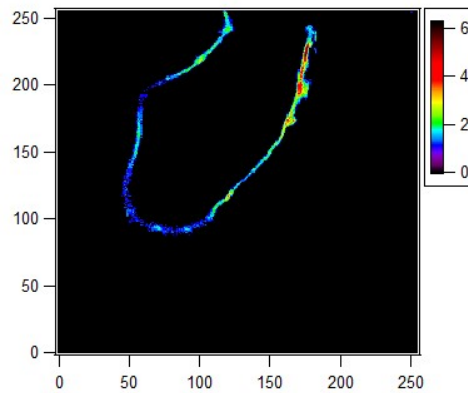
$$n = \frac{\langle F \rangle^2}{\sigma_F^2 - \langle F \rangle}$$

- Introduction
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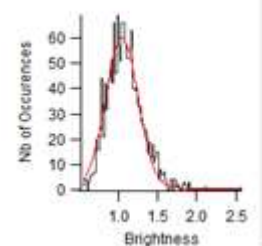
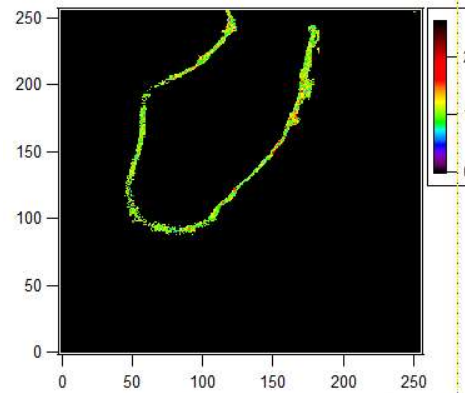
N&B : Data representations



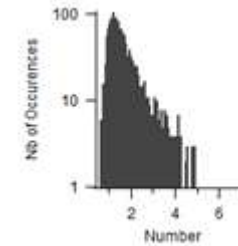
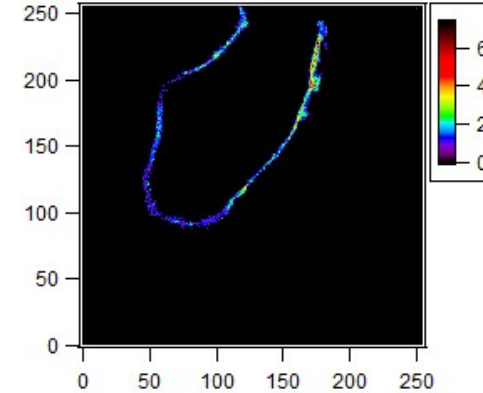
Intensity map



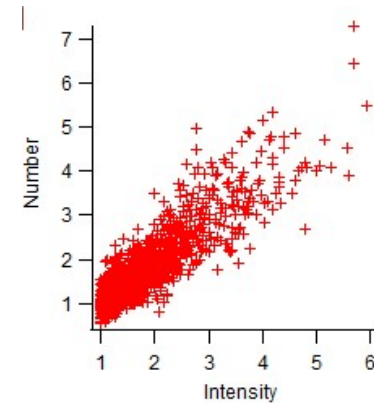
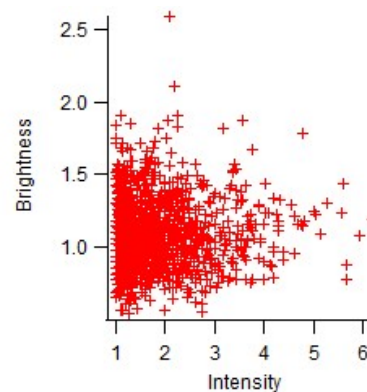
Brightness map



Number map

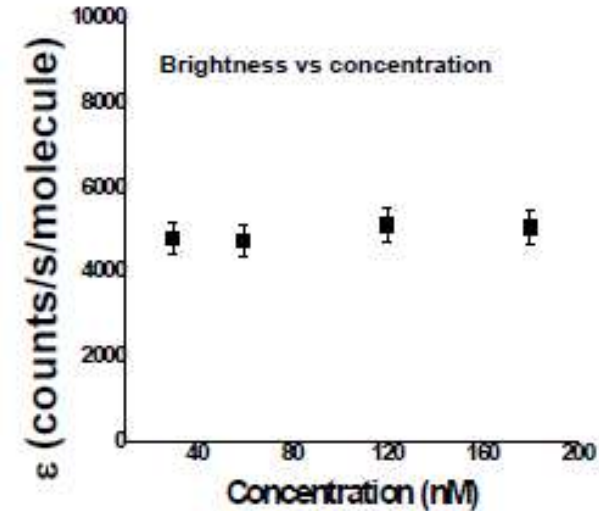
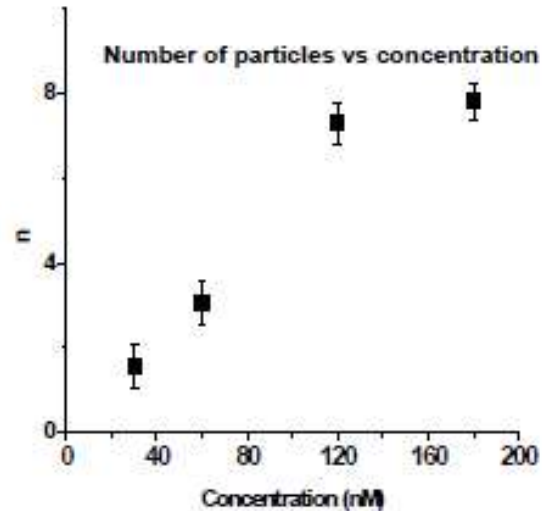
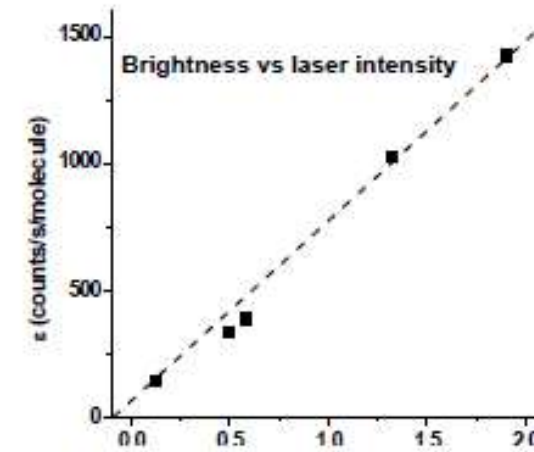
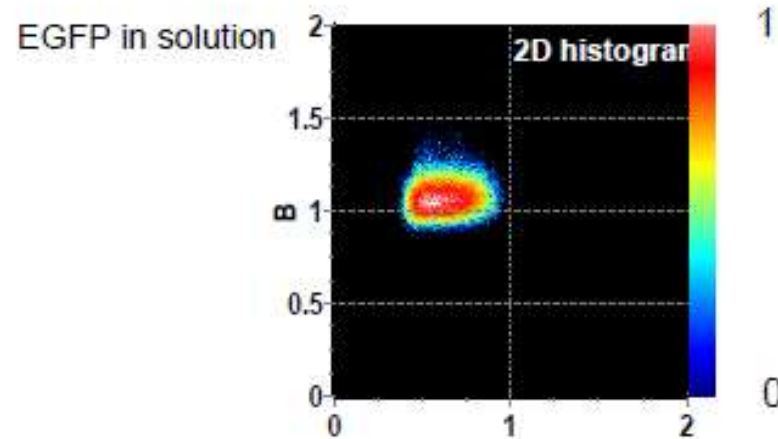
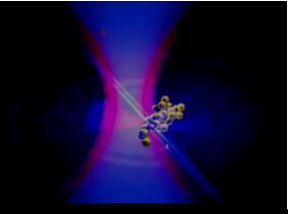


- Home-made software @CBS
- SimFCS by LFD
- Other companies (ISS, ...)



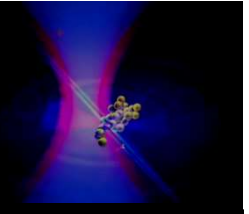
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- Cross correlation

N&B : Demonstration

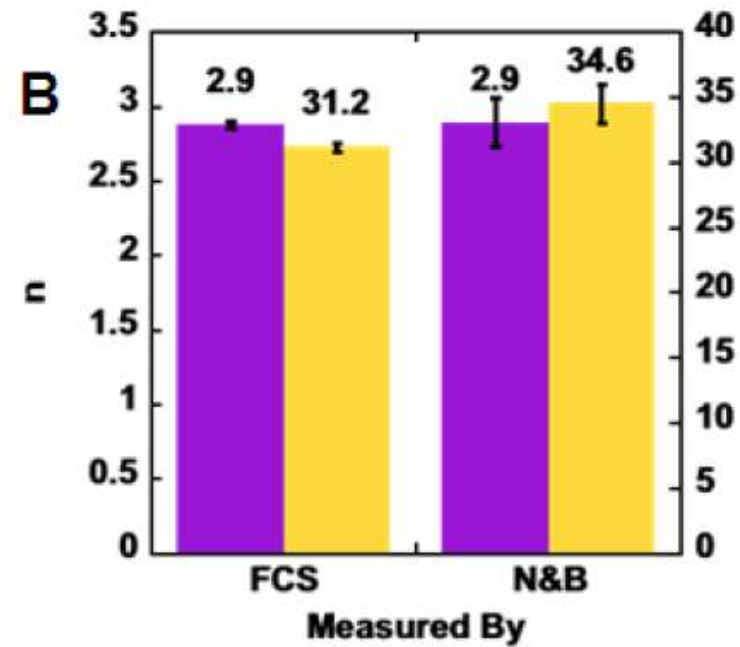
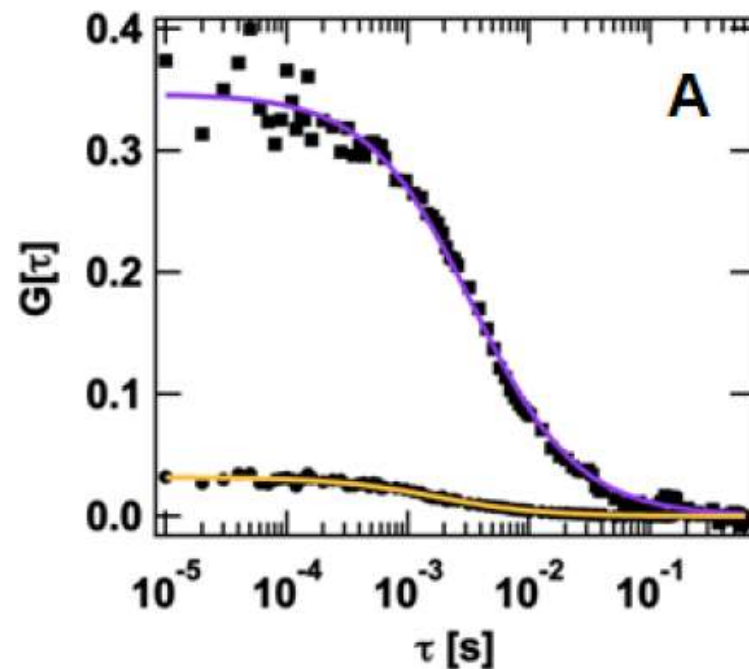


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N&B : Demonstration

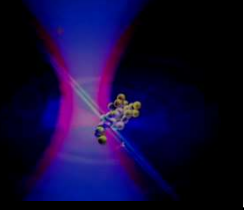


Rhodamine 110 in 75% glycerol at 16nM & 160nM



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N&B : the case of immobile particles



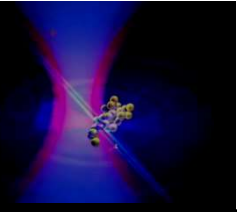
For immobile particles, there is no contribution of the number of molecules to the variance

$$B = \frac{\sigma_F^2}{\langle F \rangle} = \frac{\sigma_n^2 + \sigma_d^2}{\langle F \rangle} = \frac{\varepsilon n}{\varepsilon n} = 1$$

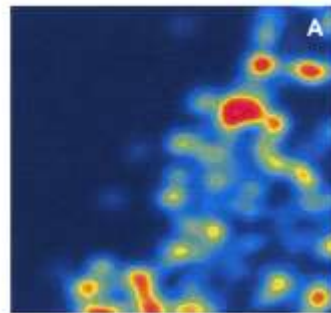
In the presence of a mixture of immobile and mobile particles in a pixel, B will have an intermediate value between 1 and the value for mobile particles.

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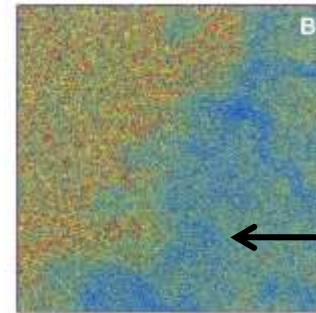
N&B : the case of immobile particles



Sample : fluorescent beads in a 100nM solution of fluorescein

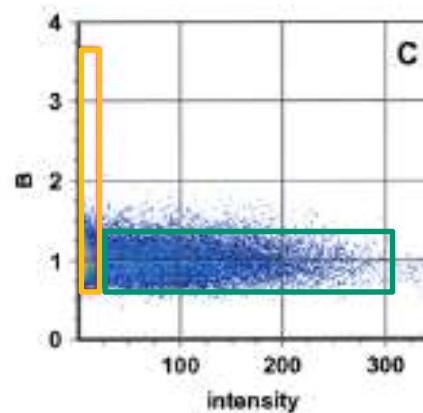


Intensity image



Brightness image

← Beads with $B=1$



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Short dwell time

$F(t)$



$F(t)$



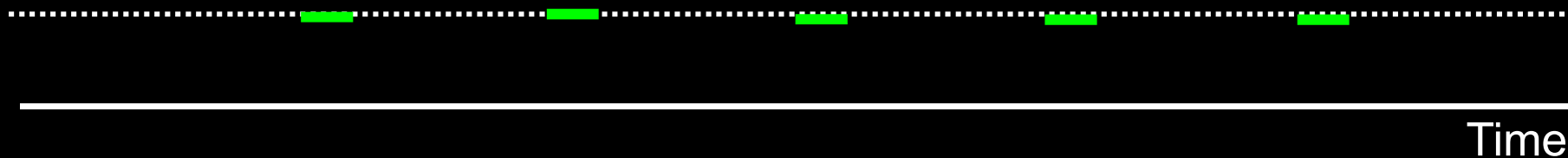
$F(t)$



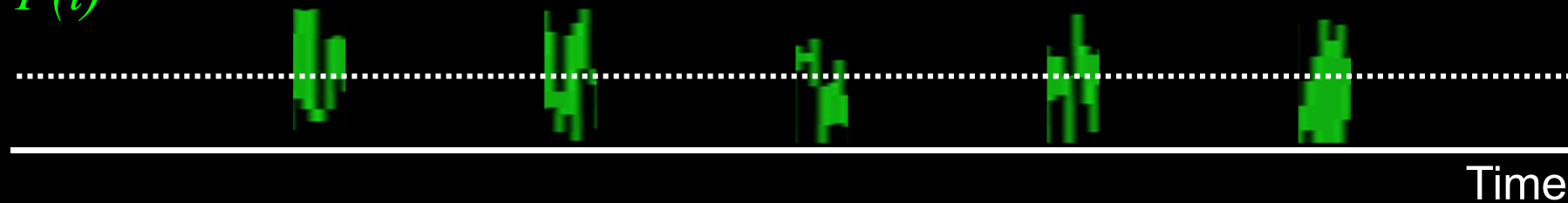
- Introduction
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Long dwell time

$F(t)$



$F(t)$

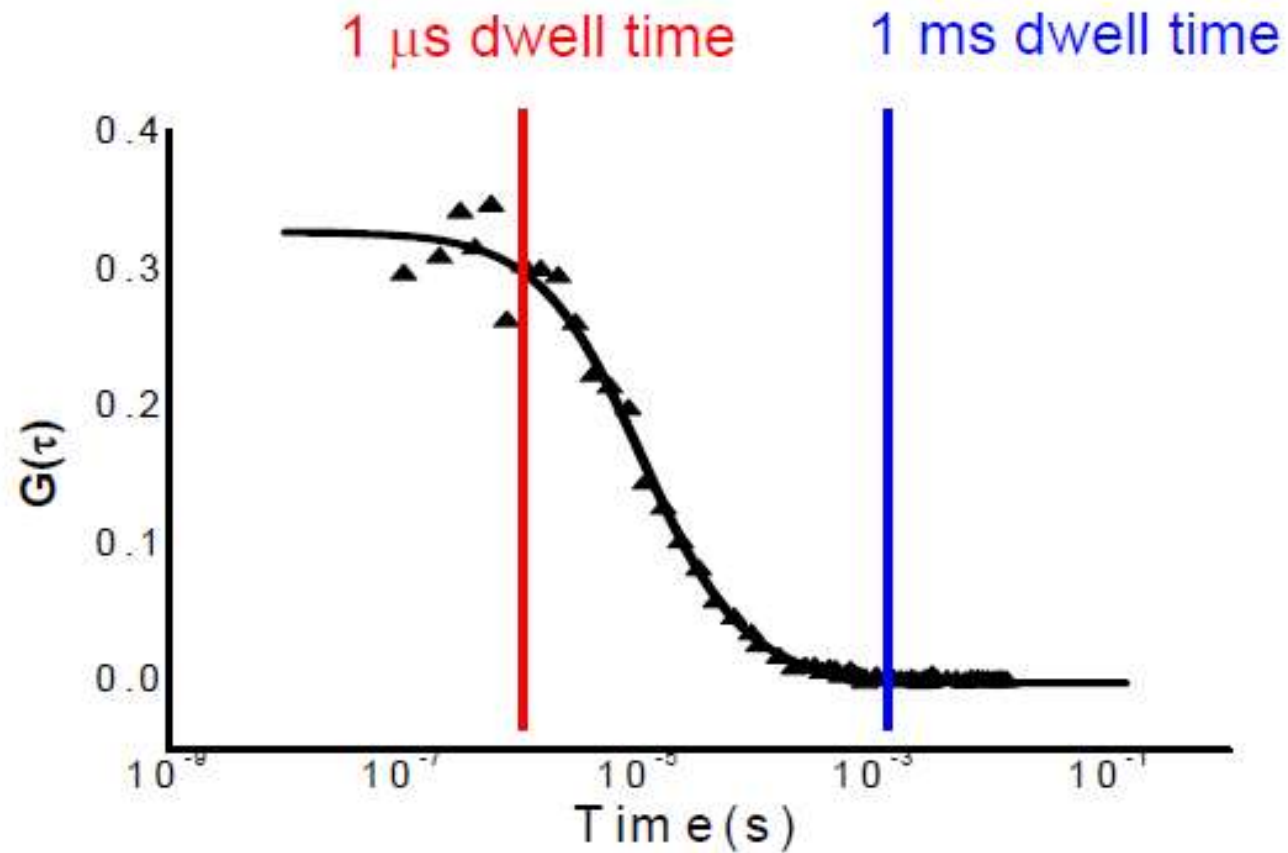
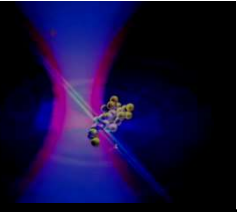


$F(t)$



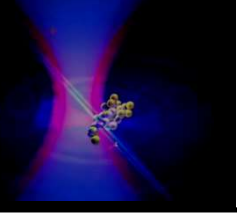
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N&B : Effect of the dwell time



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N&B summary



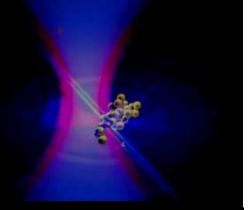
- N&B quantifies the number of molecules and their brightness for each pixel
 - A map of N&B can be obtained
- Acquisition can be done with a commercial LSM system (APD recommended)
- The immobile fraction can be detected and separated ($B=1$)
- Photobleaching of the sample is strongly reduced due to fast scanning



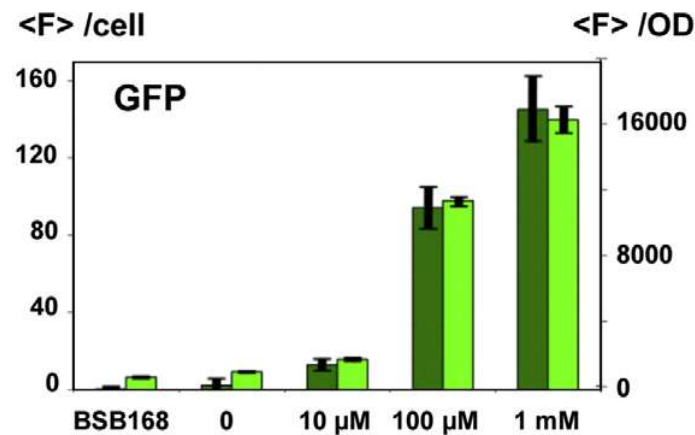
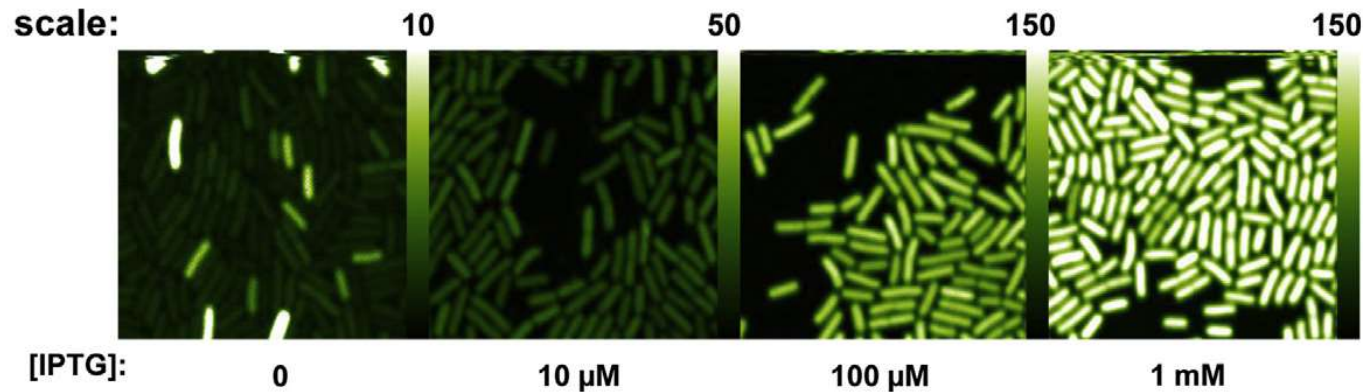
- You need fluctuations !!
 - Low concentration of fluorescent species (up to μM)
 - Low background
 - Low photobleaching

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Quantification of noise in gene expression in bacteria



Expression of Gfpmut2 under control of an inducible promoter in *B. Subtilis*



Correlation between the expression determined by 2P-microscopy and ensemble fluorescence